

APM1220: Applied Multivariate Data Analysis

Formative Assessment 1

Aldrin A. Tupas

August 27, 2026

Problem 2.2

Given the matrices

$$A = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}$$

(a) $5A$

$$5A = 5 \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 5(-1) & 5(3) \\ 5(4) & 5(2) \end{bmatrix} = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}$$

(b) BA

Matrix B has dimension 3×2 and A has dimension 2×2 . Since the inner dimensions match, BA is defined (3×2):

$$BA = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 4(-1) + (-3)(4) & 4(3) + (-3)(2) \\ 1(-1) + (-2)(4) & 1(3) + (-2)(2) \\ -2(-1) + 0(4) & -2(3) + 0(2) \end{bmatrix} = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}$$

(c) $A'B'$

By transpose properties, $A'B' = (BA)'$. Taking the transpose of BA from part (b):

$$A'B' = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}' = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}$$

(d) $C'B$

C' has dimension 1×3 and B has dimension 3×2 . The multiplication yields a 1×2 row vector:

$$\begin{aligned} C' &= [5 \quad -4 \quad 2] \\ C'B &= [5 \quad -4 \quad 2] \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix} = [5(4) + (-4)(1) + 2(-2) \quad 5(-3) + (-4)(-2) + 2(0)] \\ &= [20 - 4 - 4 \quad -15 + 8 + 0] = [12 \quad -7] \end{aligned}$$

(e) Is AB defined?

A is of dimension 2×2 and B is of dimension 3×2 . Since the number of columns of A (2) does not equal the number of rows of B (3), AB is **not defined**.

Problem 2.4

(a) **Prove** $(A')^{-1} = (A^{-1})'$

Proof: By definition of the inverse matrix, we know that:

$$AA^{-1} = I$$

Taking the transpose of both sides:

$$(AA^{-1})' = I'$$

Using the property of transpose for products $(XY)' = Y'X'$, and noting that $I' = I$:

$$(A^{-1})'A' = I$$

Similarly, starting from $A^{-1}A = I$:

$$(A^{-1}A)' = I' \implies A'(A^{-1})' = I$$

By definition, if $XY = YX = I$, then $Y = X^{-1}$. Letting $X = A'$ and $Y = (A^{-1})'$, we conclude:

$$(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$$

■

(b) **Prove** $(AB)^{-1} = B^{-1}A^{-1}$

Proof: By definition, a matrix M is the inverse of (AB) if $(AB)M = I$ and $M(AB) = I$.

Let us test $M = B^{-1}A^{-1}$: 1. Right multiplication:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

2. Left multiplication:

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

Since both products yield the identity matrix I , $B^{-1}A^{-1}$ is the unique inverse of AB :

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

■

Problem 2.8

Given the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

1. Eigenvalues λ_1, λ_2

Solve the characteristic equation $\det(A - \lambda I) = 0$:

$$\det \begin{bmatrix} 1 - \lambda & 2 \\ 2 & -2 - \lambda \end{bmatrix} = (1 - \lambda)(-2 - \lambda) - (2)(2) = 0$$

$$\lambda^2 + \lambda - 2 - 4 = 0 \implies \lambda^2 + \lambda - 6 = 0$$

$$(\lambda + 3)(\lambda - 2) = 0$$

Thus, the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -3$.

2. Normalized Eigenvectors e_1, e_2

For $\lambda_1 = 2$: Solve $(A - 2I)x = 0$:

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -x_1 + 2x_2 = 0 \implies x_1 = 2x_2$$

An eigenvector is $x^{(1)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Normalizing gives:

$$\|x^{(1)}\| = \sqrt{2^2 + 1^2} = \sqrt{5} \implies \mathbf{e}_1 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

For $\lambda_2 = -3$: Solve $(A - (-3)I)x = (A + 3I)x = 0$:

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies 2x_1 + x_2 = 0 \implies x_2 = -2x_1$$

An eigenvector is $x^{(2)} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. Normalizing gives:

$$\|x^{(2)}\| = \sqrt{1^2 + (-2)^2} = \sqrt{5} \implies \mathbf{e}_2 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{bmatrix}$$

3. Spectral Decomposition

The spectral decomposition of A is $A = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2'$:

$$e_1 e_1' = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix}$$

$$e_2 e_2' = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} \end{bmatrix}$$

Thus, the spectral decomposition is:

$$\mathbf{A} = 2 \begin{bmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix} - 3 \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} \end{bmatrix}$$

Problem 2.11

Proof by Induction: Let A be a $p \times p$ diagonal matrix:

$$A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{pp} \end{bmatrix}$$

Expanding the determinant along the first row:

$$|A| = \sum_{j=1}^p a_{1j} A_{1j}$$

where A_{1j} is the cofactor. Since $a_{1j} = 0$ for all $j \neq 1$, this simplifies to:

$$|A| = a_{11}A_{11} + 0 + \cdots + 0 = a_{11}|A^{(1)}|$$

where $A^{(1)}$ is the $(p-1) \times (p-1)$ diagonal submatrix obtained by deleting the first row and first column of A :

$$A^{(1)} = \begin{bmatrix} a_{22} & 0 & \cdots & 0 \\ 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{pp} \end{bmatrix}$$

Applying cofactor expansion along the first row of $A^{(1)}$:

$$|A^{(1)}| = a_{22}|A^{(2)}|$$

Repeating this process iteratively for $k = 1, 2, \dots, p-1$ yields:

$$|A| = a_{11} \cdot a_{22} \cdot |A^{(2)}| = a_{11} \cdot a_{22} \cdots a_{pp}$$

Thus, the determinant of a diagonal matrix is the product of its diagonal elements:

$$|\mathbf{A}| = \prod_{i=1}^p \mathbf{a}_{ii} = \mathbf{a}_{11} \mathbf{a}_{22} \cdots \mathbf{a}_{pp}$$

■

Problem 2.24

Given $\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

(a) Find Σ^{-1}

For a diagonal matrix, the inverse is obtained by taking the reciprocals of the diagonal elements:

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 \\ 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Eigenvalues and Eigenvectors of Σ

Since Σ is diagonal, its eigenvalues are the diagonal entries:

$$\lambda_1 = \mathbf{9}, \quad \lambda_2 = \mathbf{4}, \quad \lambda_3 = \mathbf{1}$$

The corresponding normalized eigenvectors are the standard basis vectors:

$$\mathbf{e}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(c) Eigenvalues and Eigenvectors of Σ^{-1}

The eigenvectors of Σ^{-1} are identical to those of Σ , and its eigenvalues are the reciprocals of the eigenvalues of Σ :

$$\lambda_1^* = \frac{1}{\mathbf{9}}, \quad \lambda_2^* = \frac{1}{\mathbf{4}}, \quad \lambda_3^* = \mathbf{1}$$

The corresponding normalized eigenvectors are:

$$\mathbf{e}_1^* = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2^* = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3^* = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Problem 2.30

Given $\mu_X = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}$, $\Sigma_X = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix}$, partitioned as:

$$X^{(1)} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad X^{(2)} = \begin{bmatrix} X_3 \\ X_4 \end{bmatrix}$$

$$\mu^{(1)} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad \mu^{(2)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \quad \Sigma_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}, \quad \Sigma_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}, \quad \Sigma_{21} = \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$$

with $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$.

(a) $E(X^{(1)})$

$$\mathbf{E}(\mathbf{X}^{(1)}) = \mu^{(1)} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

(b) $E(AX^{(1)})$

$$E(AX^{(1)}) = AE(X^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 1(4) + 2(3) = \mathbf{10}$$

(c) $Cov(X^{(1)})$

$$\mathbf{Cov}(\mathbf{X}^{(1)}) = \mathbf{\Sigma}_{11} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

(d) $Cov(AX^{(1)})$

$$Cov(AX^{(1)}) = A\Sigma_{11}A' = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3(1) + 2(2) = \mathbf{7}$$

(e) $E(X^{(2)})$

$$\mathbf{E}(\mathbf{X}^{(2)}) = \mu^{(2)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(f) $E(BX^{(2)})$

$$E(BX^{(2)}) = BE(X^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1(2) - 2(1) \\ 2(2) - 1(1) \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

(g) $Cov(X^{(2)})$

$$\mathbf{Cov}(\mathbf{X}^{(2)}) = \mathbf{\Sigma}_{22} = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}$$

(h) $Cov(BX^{(2)})$

$$\begin{aligned} Cov(BX^{(2)}) &= B\Sigma_{22}B' = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix} \end{aligned}$$

(i) $Cov(X^{(1)}, X^{(2)})$

$$\mathbf{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)}) = \mathbf{\Sigma}_{12} = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$$

(j) $Cov(AX^{(1)}, BX^{(2)})$

$$\begin{aligned} Cov(AX^{(1)}, BX^{(2)}) &= A\Sigma_{12}B' = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 4(1) + 2(-2) & 4(2) + 2(-1) \end{bmatrix} = \begin{bmatrix} 0 & 6 \end{bmatrix} \end{aligned}$$

Problem 2.41

$$\text{Given } \mu_X = \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix}, \Sigma_X = 3I_4 = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}, \text{ and } A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}.$$

(a) Find $E(AX)$

$$\begin{aligned} E(AX) &= A\mu_X = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 1(3) - 1(2) + 0 + 0 \\ 1(3) + 1(2) - 2(-2) + 0 \\ 1(3) + 1(2) + 1(-2) - 3(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix} \end{aligned}$$

(b) Find $Cov(AX)$

$$Cov(AX) = A\Sigma_X A' = A(3I_4)A' = 3AA'$$

First compute AA' :

$$AA' = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

Multiplying by 3:

$$Cov(AX) = 3 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}$$

(c) Which pairs of linear combinations have zero covariances?

Since $Cov(AX)$ is a diagonal matrix, all off-diagonal entries are zero. Therefore, **all pairs of linear combinations** have zero covariance:

- Row 1 and Row 2: $Cov((AX)_1, (AX)_2) = 0$
- Row 1 and Row 3: $Cov((AX)_1, (AX)_3) = 0$
- Row 2 and Row 3: $Cov((AX)_2, (AX)_3) = 0$